(ii) $2x^2 + 5x + 7 = 0$

$a = 2, \quad b = 5, \quad c = 7$

$$\alpha + \beta = -\frac{b}{a} = \frac{-5}{2} \text{ and } \alpha\beta = \frac{c}{a} = \frac{7}{2}$$

$$\alpha + \beta = -\frac{5}{2};\ \alpha\beta = \frac{7}{2}$$

(iii) $kx^2 - k^2x - 2k^3 = 0$

$a = k, \quad b = -k^2, \quad c = -2k^3$

$$\alpha + \beta = -\frac{b}{a} = \frac{-(-k^2)}{k} = k \text{ and } \alpha\beta = \frac{c}{a} = \frac{-2k^3}{k} = -2k^2$$

Exercise 3.9

1. Determine the quadratic equations, whose sum and product of roots are

   (i) $-9,\ 20$ (ii) $\frac{5}{3},\ 4$ (iii) $\frac{-3}{2},\ -1$ (iv) $-(2-a)^2,\ (a+5)^2$

2. Find the sum and product of the roots for each of the following quadratic equations

   (i) $x^2 + 3x - 28 = 0$ (ii) $x^2 + 3x = 0$ (iii) $3 + \frac{1}{a} = \frac{10}{a^2}$ (iv) $3y^2 - y - 4 = 0$

3.6.4 Solving Quadratic Equations

We have already learnt how to solve linear equations in one, two and three variable(s). Recall that the values of the variables which satisfies a given equation are called its solution(s). In this section, we are going to study three methods of solving quadratic equation, namely factorization method, completing the square method and using formula.

Solving a quadratic equation by factorization method.

We follow the steps provided below to solve a quadratic equation through factorization method.

Step 1: Write the equation in general form $ax^2 + bx + c = 0$

Step 2: By splitting the middle term, factorize the given equation.

Step 3: After factorizing, the given quadratic equation can be written as product of two linear factors.

Step 4: Equate each linear factor to zero and solve for x.

These values of x gives the roots of the equation.

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